

When Regional Favoritism Fades

Nightlights and Pollution in Leader Birth Regions

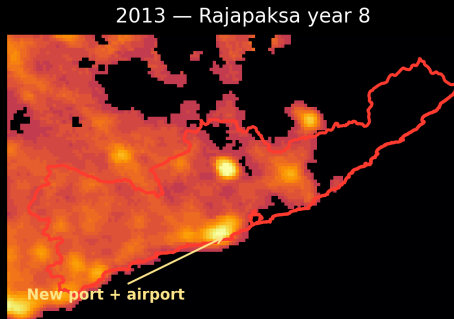
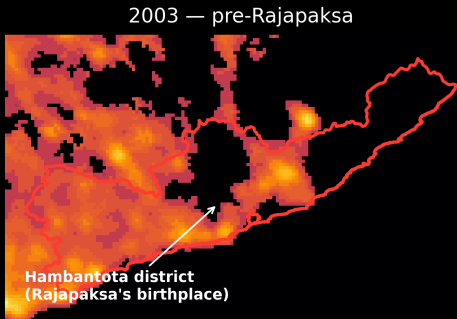
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INFO 288: Big Data and Development

Prof. Joshua Blumenstock

Hambantota, Sri Lanka at night



Rajapaksa took office in 2005. By 2013, his home district had a \$1.5B Chinese-financed port, an international airport named after him, a cricket stadium, and a conference center. Most are barely used.

Why birth regions?

- Leaders direct resources to where they're **from**
- Hodler & Raschky (*QJE*, 2014) show this globally in satellite nightlights
- The effect is largest in Sub-Saharan Africa

Our question

Does it **still hold** after 2005?

This project

Two questions:

1. Does the birth-region nightlights effect **replicate after 2005**?
2. If not, did favoritism shift from *more* growth to *cleaner* growth? (“green favoritism”)

Preview:

- Long DMSP (1992–2013): replicates ($\beta = 0.013$, $p = 0.030$)
- Post-2005: estimates collapse, and no subsample brings them back
- Pollution doesn't show cleaner growth either

Data: a global district-year panel

Inputs

- PLAD: leader birthplaces & spells
- DMSP + harmonized DMSP–VIIRS nightlights
- ACAG NO₂, ACAG PM_{2.5}
- V-Dem, GADM ADM2 boundaries

Sample

- ADM2 district $\times$ year
- Global, up to ~ 950 k obs
- Treated = current leader's birth district

$$\log(Y_{ict} + 0.01) = \beta \cdot \text{BirthRegionLeader}_{ict} + \alpha_i + \gamma_{ct} + \varepsilon_{ict}$$

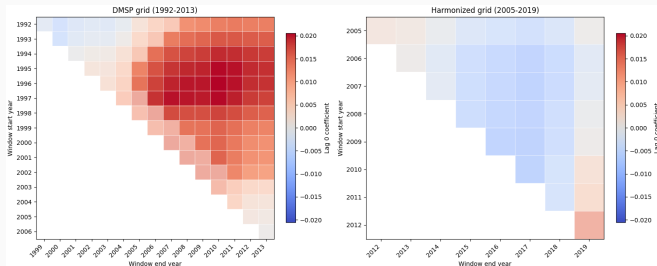
- ADM2 FE α_i , country-year FE γ_{ct}
- β identified by leader transitions within country-year
- SEs clustered by leader spell
- Outcomes: log NL, log NO₂, intensity = log NO₂ – log NL

Result 1: replication holds, post-2005 collapses

Window	Coef.	SE	<i>p</i>	N
1992–2013	0.0128	0.0059	0.030	951,915
1995–2013	0.0187	0.0055	0.001	829,988
2000–2013	0.0110	0.0058	0.056	615,660
2005–2013	0.0017	0.0058	0.768	398,636
2005–2019	0.0004	0.0087	0.966	671,399

Long DMSP: $\sim 1.3\text{--}1.9\%$ birth-region brightness premium. Post-2005: gone.

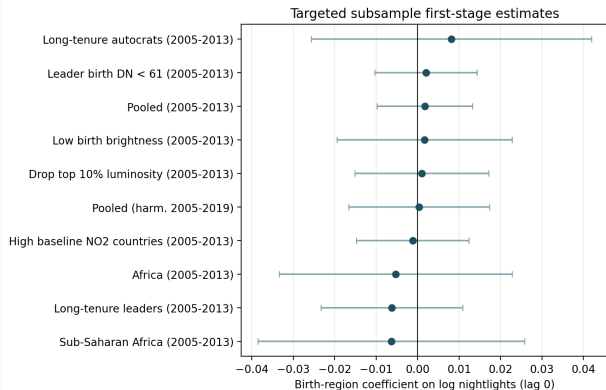
Result 2: attenuation is broad



Each cell is a separate regression.

- Positive estimates cluster in DMSP windows ending late 2000s
- Post-2005 windows: near zero everywhere

Result 3: no subsample rescues it



Searched where the effect should be strongest:

- Africa / SSA
- Long-tenure autocrats
- Low-brightness districts
- High-NO₂ countries

None passes coef. ≥ 0.005 , $p < 0.10$.

Result 4: no green favoritism

If leaders shifted from more growth to cleaner growth, pollution should *fall* in birth regions.

Outcome (2005–2019, lag 0)	Coef.	SE	<i>p</i>	N
log Nightlights	−0.003	0.009	0.778	631k
log NO ₂	+0.031	0.015	0.045	631k
NO ₂ intensity	+0.033	0.018	0.063	631k
PM _{2.5} intensity (lag 2)	+0.018	0.007	0.019	669k

Estimates go the **wrong way**. Less precise under country clustering, but never negative.

What could drive the attenuation?

- Observed spells are shorter in the later window (5.2 $\rightarrow$ 4.0 yrs). Partly a mechanical artifact of window truncation.
- Birth districts are already brighter, closer to DMSP saturation. But dropping bright districts doesn't help.
- Not harmonization: VIIRS-only 2018–2023 is also ≈ 0
- Not worse PLAD geocoding: quality is stable across periods
- Green favoritism would predict falling pollution. We don't see that.

Bottom line

The signal is gone, and the easy explanations don't hold.

1. The long DMSP result **still replicates**
2. After 2005 the effect collapses, even where theory says it should be strongest
3. Pollution outcomes are zero-to-positive, so it isn't green favoritism
4. A textbook political-economy result looks much weaker once you push it past 2005

Thank you!

`github.com/RichSchulz/political-leader-pollution`

Appendix: subsample activity estimates

Sample	Window	Coef.	SE	<i>p</i>
Africa	DMSP 1992–2013	0.0428	0.0163	0.009
Sub-Saharan Africa	DMSP 1992–2013	0.0442	0.0174	0.011
Non-SSA	DMSP 1992–2013	0.0053	0.0059	0.370
Africa	DMSP 2005–2013	−0.0052	0.0143	0.714
Sub-Saharan Africa	DMSP 2005–2013	−0.0063	0.0164	0.700
Long-tenure leaders	DMSP 2005–2013	−0.0062	0.0087	0.479
Long-tenure autocrats	DMSP 2005–2013	0.0081	0.0173	0.637
Drop top 10% luminosity	DMSP 2005–2013	0.0010	0.0082	0.899
Low birth brightness	DMSP 2005–2013	0.0016	0.0108	0.880
Africa	Harmonized 2005–2019	0.0262	0.0189	0.167
Sub-Saharan Africa	Harmonized 2005–2019	0.0315	0.0210	0.135